

AI FitTrack

8. CONFIGURE MONGODB

8.1 Introduction

MongoDB is a NoSQL database used to store and manage the data of the AI FitTrack application. Unlike traditional relational databases, MongoDB stores data in flexible JSON-like documents. MongoDB can be used to store user profiles, fitness information, BMI results, workout plans, diet plans, calorie records, and progress details.

8.2 MongoDB Installation

The following steps are used to configure MongoDB for the AI FitTrack project:

1. Download and install MongoDB Community Server.
2. Install MongoDB Compass for graphical database management.
3. Start the MongoDB service.
4. Open MongoDB Compass.
5. Connect to the local MongoDB server.

Default local connection:

```
mongodb://localhost:27017
```

8.3 Create Database

Create a database named:

```
AI_FitTrack
```

MongoDB automatically creates the database when the first document is inserted.

8.4 Database Collections

The following collections can be created for the AI FitTrack system:

```
AI_FitTrack
■
■■■ users
■■■ bmi_records
■■■ calories
■■■ workouts
■■■ diets
■■■ progress
```

Users Collection

Stores user registration and profile information.

```
{
  "name": "Charlie",
  "email": "user@example.com",
  "password": "hashed_password",
  "age": 22,
  "gender": "Male",
  "height": 170,
  "weight": 65
}
```

BMI Records Collection

Stores BMI calculation results.

```
{
  "user_id": "USER_ID",
  "height": 170,
  "weight": 65,
  "bmi": 22.49,
  "category": "Normal Weight",
  "date": "2026-09-23"
}
```

Workout Collection

Stores recommended or completed workout information.

```
{
  "user_id": "USER_ID",
  "exercise": "Running",
  "duration": 30,
  "calories_burned": 250,
  "date": "2026-09-23"
}
```

Diet Collection

Stores diet and meal information.

```
{
  "user_id": "USER_ID",
  "meal_type": "Breakfast",
  "food": "Oatmeal",
  "calories": 300,
  "date": "2026-09-23"
}
```

Progress Collection

Stores fitness progress information.

```
{
  "user_id": "USER_ID",
  "weight": 64,
  "bmi": 22.15,
  "date": "2026-09-23"
}
```

8.5 MongoDB Connection

For a PHP-based application, MongoDB can be connected using the MongoDB PHP extension and Composer MongoDB library.

Install the MongoDB PHP library:

```
composer require mongodb/mongodb
```

Example PHP connection:

```
<?php

require 'vendor/autoload.php';

$client = new MongoClient(
```

```

        "mongodb://localhost:27017"
    );

    $database = $client->AI_FitTrack;

    echo "MongoDB Connected Successfully";

    ?>

```

8.6 Insert User Data

Example PHP code for inserting a user:

```

<?php

require 'vendor/autoload.php';

$client = new MongoDB\Client("mongodb://localhost:27017");
$db = $client->AI_FitTrack;
$users = $db->users;

$user = [
    "name" => "Charlie",
    "email" => "user@example.com",
    "age" => 22,
    "gender" => "Male",
    "height" => 170,
    "weight" => 65
];

$result = $users->insertOne($user);

echo "User inserted successfully";

?>

```

8.7 MongoDB Architecture

```

AI_FitTrack
↓
PHP_Backend
↓
MongoDB_PHP_Driver
↓
MongoDB_Server
↓
Users | BMI | Workouts | Diet | Progress

```

8.8 Advantages of MongoDB

- Flexible document-based data storage.
- Easy to handle changing fitness data.
- Supports large amounts of user data.
- Fast data retrieval for suitable workloads.
- JSON-like documents are easy to work with in web applications.
- Suitable for storing different types of fitness records.
- Can be scaled as the application grows.

8.9 Conclusion

MongoDB provides a flexible database solution for AI FitTrack. It can store user profiles, BMI results, calorie information, workout details, diet plans, and fitness progress in document format. The PHP backend communicates with MongoDB to create, read, update, and delete fitness-related data.